

Visual Layer

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Introduction

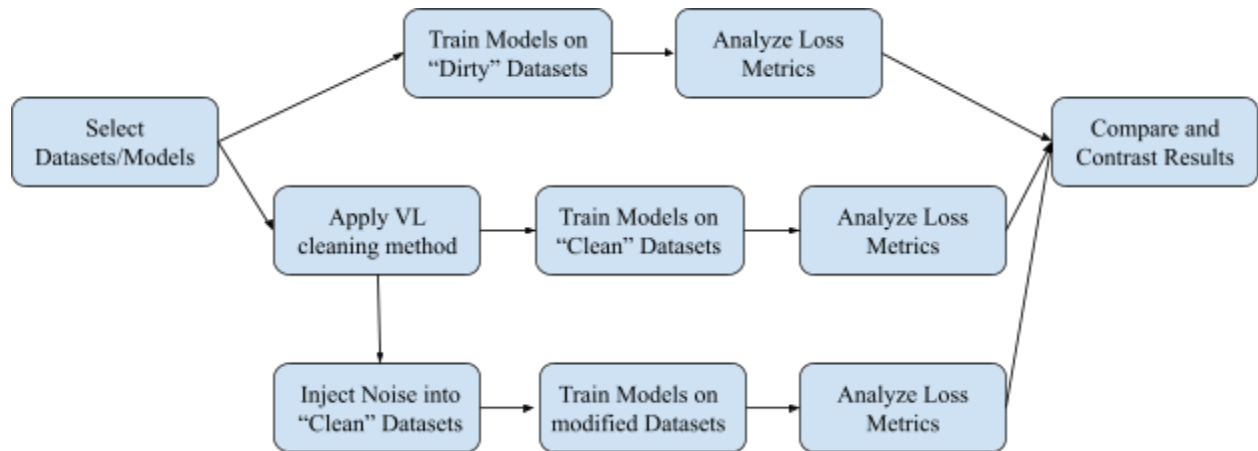
Problem

Despite the rapid advancement of computer vision models, dataset quality remains one of the most under-quantified determinants of model performance. Prior studies have shown that even modest data degradation can have measurable effects. For example, image blur alone can reduce classification accuracy by 20-40%, and removing just 5-10% of low-quality or mislabeled samples can improve accuracy by 3-7%. However, most large-scale pre-training datasets like ImageNet and COCO contain 10-25% noisy or low-quality samples, ranging from mislabeled images to compression artifacts and poor lighting conditions. We aim to empirically evaluate how dataset quality affects deep learning performance in computer vision. Models will be pretrained on both raw and curated datasets to measure differences in accuracy, robustness, and generalization. Our goal is to quantify the direct impact of data cleanliness and enrichment on model outcomes.

Innovation

We will be investigating how the quality of pretraining data directly influences model accuracy, robustness, and generalizability, and how data leakage occurs in training large language models for computer vision depending on the pretraining data. We aim to develop new methods to prevent/fix data leakage in these visual dataset through systematic experiments. Hence, it is important to focus on finding a generalized method of how the quality of the pretraining data affects the quality of the machine learning model performance and which type of datasets gives the best results for training machine learning models in computer vision.

System Architecture



We will be training a small and large version each of ViT-B and ResNet on 2 datasets, ImageNet and COCO - locally first, to get familiar with the usage of both, followed by subsequent larger training runs on an Nvidia H100.

Requirements

Functional Requirements

- FR1 - The research should test our hypothesis of how the quality of the pretraining dataset directly influences model accuracy.
- FR2 - The research should quantify the impact of data cleanliness and enrichment on model performance.
- FR3 - The research should collect and organize publicly available datasets and analyze them for quality issues such as noise, mislabeling, and bias using Visual Layer's tools.
- FR4 - The research should train multiple deep learning models on both raw and curated datasets and record their performance metrics for comparison.

Non-functional Requirements

- Performance: Evidence from our experiments should be organized and analyzed on how noise and mislabeling affects the performance of the model and should show statistically significant evidence that proves or disproves our hypothesis. Our model should be able to process images quickly and efficiently with the correct labels.
- Reproducibility: Experiments should be able to be reproduced and should have a low variance within 5.

- Reliability: Experiments should be able to be rerun and data should be saved from these experiments.
- Usability: All our experiments and data collected are documented for our research paper
- Security: Licensing of visual datasets and CV models.
- Availability: The research will be open sourced and published to the ML/computer science community in regards to CV and AI.

User Stories/Use Cases

- **User Story 1: Presenting Results**

As someone trying to present or view the results of this experiment, I want to easily see differences in key performance metrics/KPIs, so that I can quickly compare and contrast the efficacy of different methods.

- **User Story 2: Reproduction of Results**

As someone trying to reproduce the results of this experiment, I want to just type a few quick commands that run the key processes of the experiment, so that I don't spend more time than I need to on reproducing the results.

- **User Story 3: Dataset Cleaning**

As someone trying to clean the dataset of this experiment, I want to identify and remove mislabeled, duplicate and low-quality images from large visual datasets, so that training data used for model development is consistent and improved model accuracy.

- **User Story 4: Image Classification**

As a military analyst trying to classify images, I want to ensure that my pre-training data will give the most accurate results and have low-noise for the AI system to correctly identify objects that are of concern.

- **Use Case 1**

- Actor: Analyst
- Generating graphs and charts tracking KPIs for model trained on clean vs. dirty data
- Sort by KPIs that are meaningfully different between clean and dirty models

- **Use Case 2**

- Actor: ML Engineer
- Exporting cleaned datasets to use for model training

- **Use Case 3**

- Actor: Data scientist
- Running automated scripts to execute each step of the experiment
- Changing script parameters to suit a modified experiment
 - Slotting in different models, datasets, etc.

- Highlight the dirty data (misabeled, duplicate, low-quality images)
- **Use Case 4**
 - Actor: Researcher
 - Goal: Wants to compare model robustness across a variety of datasets
 - Runs controlled experiments to evaluate how models trained on clean vs unclean datasets perform
 - Steps: Select model and dataset, apply noise, run experiment & evaluate
- **Use Case 5**
 - Actor: Military Analyst
 - Wants to correctly identify images that are potential threats or concerns.
 - Runs pre training data with pre-select noise to evaluate the accuracy of the image classification.
- **Use Case 6**
 - Actor: Open-Source Contributor
 - Goal: Publish curated datasets and model artifacts
 - Prepares cleaned datasets, documentation, and model outputs, then publishes them

Appendices/Glossary

- ML - **Machine Learning**
- AI - **Artificial Intelligence**
- CV - **Computer Vision**
- KPI - **Key Performance Indicator**